

# Sachin Avutu

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## Education

### Drexel University

*Bachelor of Science in Computer Science, Minor in Linguistics*

*Sep 2024 – Expected Jun 2029*

## Experience

### Full-stack Developer

*The Triangle - Drexel University Student Newspaper*

*Philadelphia, PA*

*May 2025 – Present*

- Leading migration from WordPress to a custom CMS and database architecture, enabling scalable publishing and full platform ownership
- Built a headless CMS in Go, replacing WordPress and enabling scalable content management via a REST API
- Constructed a Python ETL pipeline to migrate 11+ years of WordPress data (18k+ articles, 5k+ authors) into MariaDB, preserving data continuity despite upstream failures
- Developed a MinHash-based entity resolution system using NumPy and Numba, reducing duplicate records and improving dataset integrity in production

### Software Engineer Co-op

*eMoney Advisor*

*Radnor, PA*

*Sep 2025 – Mar 2026*

- Delivered production features and resolved technical debt in a 1.3M+ LOC .NET codebase supporting integrations with 12,000+ financial institutions
- Partnered with senior engineers to develop and deploy eMoney's Rocket Mortgage integration into production
- Implemented NUnit-based test suites to catch regressions and stabilize production release, while following TDD practices
- Traced and corrected a partner API change that triggered a spike in client-facing errors using Elasticsearch and Kibana
- Orchestrated production deployments through Jenkins and Octopus within CI/CD pipelines

## Projects

### Minhashlib

[github.com/ssavutu/minhashlib](https://github.com/ssavutu/minhashlib) 

- Released an MIT-licensed high-performance Minhash similarity detection Python library for use on large sets of strings
- Developed a high-performance MinHash library achieving 6x speedup over datasketch on CPU benchmarks while maintaining comparable accuracy
- Accelerated signature generation through vectorization and JIT compilation using Numba
- Integrated Minhashlib in a production-critical ETL pipeline at The Triangle

### EagleEyePhilly

[github.com/ssavutu/EagleEyePhilly](https://github.com/ssavutu/EagleEyePhilly) 

- Built a FastAPI service backed by Selenium automation during a hackathon to expose Philadelphia 311 workflows through a programmable API
- Deployed a demo-ready backend from a local machine using a Cloudflare Tunnel, enabling live end-to-end system demonstrations
- Authored GitHub Wiki API documentation and coached teammates on HTTP request design, testing, and debugging
- Partnered on Gemini integration to support one-shot situational analysis and resolved cross-team backend integration issues

### DragonFuel

[github.com/ssavutu/DragonFuel](https://github.com/ssavutu/DragonFuel) 

- Created a multipage nutrition tracking app using the Flutter UI framework
- Assembled a Python scraping pipeline using Selenium and BeautifulSoup to aggregate nutritional data
- Designed a local data layer using SQLite and JSON to support persistent, real-time application state

### RxBuddy

[github.com/ssavutu/RxBuddy](https://github.com/ssavutu/RxBuddy) 

- Prototyped a Flutter based medication reminder app MVP within a 24-hour hackathon
- Exposed a RESTful API via Flask enabling real-time image and JSON data exchange
- Leveraged Gemini API to extract and interpret text from prescription labels

## Skills

**Languages:** Python, C#, C++, Go, Java, JavaScript, TypeScript, Dart, SQL, HTML, CSS

**Technologies:** .NET, NUnit, MariaDB, NumPy, Numba, Flask, Flutter, React, Selenium, BeautifulSoup, SQLite, Pydantic

**Tools:** Linux, Docker, Git, GitHub, Bitbucket, Jira, Confluence, Octopus, Jenkins, Elasticsearch